

**The Role of Emotional Non-Acceptance in Mediating Encapsulated Beliefs and Depressive
Symptoms**

Niklas Kors

Student Number: 14260603

University of Amsterdam

Supervised by: Sophie Rameckers

Number of Words: 4988

31st of January 2025

Table of Contents

Abstract.....	3
Introduction.....	4
Methods.....	8
Participants.....	8
Materials	9
Procedure	12
Analytic plan.....	12
Results.....	14
Preliminary analysis.....	14
Mediation analysis	16
Sensitivity analysis.....	17
Exploratory analysis.....	18
Discussion.....	19
References.....	24
Appendix A.....	29

Abstract

Depression is a prevalent mental health disorder, often linked to cognitive distortions and emotional dysregulation. While core beliefs have been well-documented as contributing factors, encapsulated beliefs—rigid, trauma-associated beliefs tied to autobiographical memory—offer a new perspective on depression’s cognitive underpinnings. This study examines whether emotional non-acceptance mediates the relationship between encapsulated belief strength and depressive symptoms. A cross-sectional study was conducted with 211 healthy participants, who completed measures assessing the strength of an encapsulated belief, emotional non-acceptance, and depressive symptoms. Contrary to expectations, the believability of an encapsulated belief did not significantly predict depressive symptoms, nor was it mediated by emotional non-acceptance. However, an exploratory analysis using the negativity of an encapsulated belief—rather than its believability—revealed a significant partial mediation effect, suggesting that the perceived emotional valence of these beliefs may be more relevant than conviction in their role in depression. These findings highlight the complexity of cognitive and emotional factors in depression. Therefore, the relationship between encapsulated beliefs and depression should be further explored in both non-clinical and clinical populations in future studies.

The Role of Emotional Non-Acceptance in Mediating Encapsulated Beliefs and Depressive Symptoms

Depression remains one of the most prevalent and debilitating mental health conditions globally, affecting over 280 million people worldwide (World Health Organization, 2021). Characterised by persistent sadness, loss of interest in previously enjoyed activities, and cognitive and physical symptoms, depression significantly impairs individuals' quality of life and poses a substantial burden on healthcare systems (American Psychiatric Association, 2013). The underlying mechanisms of depression and the influence of related constructs are not yet fully understood. Gaining deeper insights into these mechanisms could help explain why some individuals are more vulnerable to depression and identify new target areas for treatment.

Negative life events are a known risk factor in the development of depression (Kraaij et al., 2002). In some cases, these events can be so severe that they directly contribute to disorders such as PTSD (McLaughlin & Lambert, 2017). However, in the case of depression, it is not only the occurrence of the event that plays a role, but, more importantly, the appraisal and interpretation of the event (A. T. Beck & Bredemeier, 2016). Such interpretations may be reinforced by similar events, forming deeply held cognitive schemas known as core beliefs. These beliefs are formally described as deep-seated assumptions about oneself, others, or the world (J. S. Beck, 2011). Furthermore, these beliefs are often rigid views that could reinforce feelings of worthlessness, hopelessness, and helplessness (A. T. Beck, 1979). The association between core beliefs and depressive symptoms has been well established by prior studies. Osmo et al. (2018) conducted a study to validate the Negative Core Beliefs Inventory (NCBI), a measure designed to assess core beliefs as outlined in Beck's cognitive theory (Leahy, 1979). Using a large sample of participants, their findings demonstrated that negative core beliefs, particularly those related to worthlessness and helplessness, are strongly

associated with depressive symptoms. Consistently, studies established by Dozois & Rnic (2015) and Cowan et al. (2019) also found that stronger negative core beliefs and fewer positive core beliefs are linked to depressive symptoms in both clinical and non-clinical youth populations.

While core beliefs represent generalized assumptions about the self, others, and the world, a related but distinct construct has emerged in recent research, namely encapsulated beliefs. Unlike core beliefs, which develop through repeated experiences and reinforce overarching cognitive schemas, encapsulated beliefs are directly tied to specific autobiographical memories and are activated along with the trauma memory itself (Rameckers et al., 2024). These beliefs often carry emotional salience linked to the original experience, which may contribute to their persistence over time. This distinction is important in understanding the persistence of depressive symptoms, as encapsulated beliefs may function as isolated, yet potent, cognitive structures that sustain distress and reinforce maladaptive emotional responses.

The association between encapsulated beliefs and depression can be explained through the theory of Emotional Schema Therapy (EST; Leahy, 2022). EST expands Beck's cognitive theory by addressing the emotional schemas. These emotional schemas, often referred to as meta-emotional problems, represent how people perceive and evaluate their emotional experiences (Ellis, 2003). For example, a person with depression may not only think, "I am worthless," but may also engage in a meta-emotional thought, such as: "I shouldn't feel this way". This latter thought reflects a non-accepting emotional stance, which has been shown to correlate with depression (Yoon et al., 2018). Consequently, individuals with depressive symptoms may struggle not only with negative cognitions but it may also indicate difficulties in regulating emotions. By introducing emotional schemas, EST provides a more elaborate explanation for why encapsulated beliefs are associated with depressive

symptoms. Furthermore, it highlights how approaches that address both cognition and emotional processing, such as imagery rescripting therapy (ImRs), may be effective. Indeed, the structured framework of ImRs, as outlined by Arntz & Weertman (1999), shows that ImRs alleviates distress by modifying the underlying schema as well as the emotional responses tied to trauma.

Emotional non-acceptance, which is one of the facets of emotional regulation, can be conceptualised as an emotional schema within EST, since it represents a maladaptive belief system about the unacceptability of one's emotions (Leahy, 2022). This schema fosters a rejection or resistance to emotional experiences, leading individuals to struggle with acknowledging or tolerating their emotions (Gratz & Roemer, 2004). The adoption of such a schema could partially explain why encapsulated beliefs can influence depressive symptoms. This schema is relevant, especially in affective disorders such as depression, where it contributes to depressive rumination, a repetitive focus on negative thoughts and feelings tied to depressive symptoms (Nolen-Hoeksema, 2000). By inhibiting emotional processing, this schema could not only exacerbate depressive symptoms but could also sustain them over time.

The relationship between the strength of an encapsulated belief and emotional non-acceptance has not been previously studied. In this research, the strength of a negative encapsulated belief is defined by its believability. Because these beliefs are rooted in negative emotional memories of past experiences, they may persist despite being irrational. Their persistence suggests a form of cognitive and emotional inflexibility, which may conceptually overlap with emotional non-acceptance, as both reflect a rigid stance toward one's thoughts and feelings. Notably, cognitive and emotional inflexibility have been identified as core factors in depression (Stange et al., 2017). Therefore, it is reasonable to hypothesize that

individuals with high levels of emotional non-acceptance are more likely to hold rigid negative beliefs.

A strong tendency to reject or perceive one's own emotional experiences as bad or unacceptable has been consistently linked to both subclinical depressive symptoms and clinically diagnosed depressive disorders (Bakhshaie et al., 2014; Brockmeyer et al. 2012). The effects of emotional non-acceptance on depression have also been explored in a study by Flynn et al. (2010) which compared the effects across genders. Lower acceptance of emotions was linked to higher depressive symptoms in both genders. Similarly, a meta-analytic review by Yoon et al. (2018) found a strong association between emotional non-acceptance and depression. This study concluded that individuals who struggle to accept their emotions tend to experience higher levels of depression, regardless of whether they are in clinical or non-clinical populations.

Despite the fact that some of the evidence and theory suggest that emotional non-acceptance plays a role in linking encapsulated beliefs to depressive symptoms, its specific function as a mediator in this relationship has not been studied. However, a study by Becerra et al. (2024) did investigate emotion regulation as a mediator in the relationship between beliefs about the controllability of emotions and depression. This study demonstrated that beliefs about the controllability of emotions are linked to difficulties in emotion regulation, which, in turn, mediates the relationship with depressive and anxiety symptoms, as well as broader psychopathological outcomes. While this provides evidence for the association between beliefs on emotional regulation and depression, it does so only for beliefs about emotional controllability and emotion regulation in general.

The presented studies provide partial evidence for the influence of encapsulated beliefs and emotional non-acceptance on depressive symptoms. The EST model provides a theoretical framework for explaining the relationship between these constructs. However, this

particular model has not been investigated in previous research. And while previous studies have examined encapsulated beliefs primarily in the context of treatment outcomes, this study extends the investigation to their role in depression directly. Therefore, the primary aim of the present study is to examine the extent to which the strength of encapsulated beliefs predicts depressive symptoms in non-clinical populations, and how this effect might be mediated by emotional non-acceptance. Two dimensions of encapsulated beliefs will be explored: their believability and their negativity at the time of measurement. In the main analysis, encapsulated beliefs will be operationalized through their believability. However, some individuals may perceive their beliefs as highly believable yet not particularly distressing, while this emotional connection may be crucial in its relation to emotional non-acceptance. To further explore the construct of encapsulated beliefs, an exploratory analysis will be conducted by using the negativity of an encapsulated belief instead of its believability. Based on the presented theoretical model and previous findings, it is hypothesized that emotional non-acceptance partially mediates the relationship between the strength of an encapsulated belief on depressive symptoms. Secondly, it is also hypothesised that when correcting for emotional non-acceptance there is a significant direct effect of encapsulated beliefs strength on depressive symptoms. This is expected because, besides the hypothesized mediation of emotional non-acceptance, other factors are still expected to play a role in the relationship between the strength of an encapsulated belief and depressive symptoms.

Methods

Participants

Participants ($N = 211$) included 159 women (75.36%), 47 men (22.27%), and 5 individuals of other or unspecified gender (2.37%). The most common nationalities were Dutch (32.7%), Greek (13.3%) and Indian (5.21%). The mean age of the participants was

26.07 years ($SD = 12.52$). All participants were recruited through online channels and the study was advertised through the research lab's website of the University of Amsterdam. All participants who volunteered to take part in the study were rewarded with either a voucher or research participation credits. Inclusion criteria required participants to be at least 18 years old and be proficient in speaking English.

Sample size determination

The minimum required sample size for the mediation analysis was determined by using an a priori power analysis based on Monte Carlo simulations (Shoemann et al., 2024). For a target power of 0.8, an alpha level of .05, and standardized path coefficients of $a = 0.2$ (small- moderate effect), $b = 0.3$ (moderate effect) and $c' = 0.1$ (small effect), the analysis indicated that a sample size exceeding 196 participants would meet the requirements of this study. There is no strong direct evidence for path a , yet it is hypothesised that there is an association with encapsulated belief strength, as both constructs might stem from similarly rigid cognitive pattern. Therefore, a small-to-moderate effect is chosen for path a . The moderate effect for path b is supported by prior findings from Yoon et al. (2018). Finally, a small effect for the direct path c' was chosen based on the findings of Osmo et al. (2018), and because a direct effect is still anticipated after correcting for the mediator.

Materials

This study utilized an online questionnaire, delivered through Qualtrics' online application (Qualtrics, 2025). The questionnaire presented questions to assess the strength of an encapsulated belief, emotion non-acceptance strategies and depressive symptoms. The questionnaire included other questionnaires as well but for the purpose of this study only the following were relevant for measuring the constructs: the Emotional Memory Network Assessment (EMNA; Philippe et al., 2009) for measuring the strength of an encapsulated

belief, the Difficulties in Emotion Regulation Scale - Short Form (DERS-SF; Kaufman et al., 2016) to measure emotion regulation non-acceptance strategies and the Patient Health Questionnaire (PHQ-9; Kroenke et al., 1999) for measuring the severity of depressive symptoms.

The EMNA questionnaire

The EMNA is based on the existing questionnaires from Philippe et al. (2009) and also includes a set of items based on the Memory Experiences Questionnaire Short Form by Luchetti & Sutin (2016). The EMNA prompts participants to recall a negative memory from childhood or adolescence and describe it in a few sentences (e.g. “being bullied” or “feeling unloved”). It continues by letting the participant specify different characteristics that included: the age of the participant in the memory, the type of event (e.g. Emotional abuse, Serious injury, Sexual abuse), and the perspective (e.g. “It happened to me directly”, “I witnessed it”, “I learned about it..”). Next, the participant rated different characteristics related to the memory including intrusiveness, emotionality, vividness and the strength of various emotions, that were all rated from 0 (not at all) to 100 (strongly agree). Following was a set of 16 questions related to emotional needs that are related to the particular memory. These questions were all rated on a 7-point Likert scale ranging from 1 (fully disagree) to 7 (fully agree). Finally, the participant was asked to formulate an encapsulated belief related to the memory that says something about themselves, others, or the world (e.g. “I am powerless”, “I am guilty”) and rate how convincing this belief is at the present moment, rating it from 0 (not at all convincing) to 100 (extremely convincing). Besides this, the participants also rated the belief in terms of negativity from 0 to a 100. After completing all questions for a particular memory, the procedure was repeated by presenting the same questions for an additional related memory that came to mind. However, responses to these

additional memories were not considered in this study. It is also important to note that this questionnaire has not yet undergone formal validation.

The DERS-SF

The DERS-SF, a validated 18-item measure by Kaufman et al. (2016), was used to assess emotion regulation difficulties. It included six sub-scales: strategies, non-acceptance, impulse, goals, awareness and clarity. Each item was rated from 1 (almost never) to 5 (almost always). Only the non-acceptance subscale is relevant for this study. An example question that was used in this subscale was: “When I’m upset, I become embarrassed for feeling that way”. Each subscale contained three dedicated questions and the sum of the scores indicated difficulty in emotion regulation. The minimum score for the subscale is 3 and the maximum score is 15. Higher scores indicate in the case of non-acceptance, a decreased ability to accept one’s emotions. This measure has been validated on a non-clinical population by Danasmita et al. (2024) and revealed an overall good internal consistency with a Cronbach’s alpha of .89 and moderate test-retest reliability.

The PHQ-9

The PHQ-9 is a validated 9-item measure by Kroenke et al. (1999), used to measure the severity of depressive symptoms. Each item in the questionnaire was answered in terms of frequency of depressive symptoms in the participant’s last two weeks prior to the study on a 4-point Likert scale from 1 (not at all) to 4 (nearly every day). An example item from the questionnaire was: “Little interest or pleasure in doing things”. By summing each item, a minimum score of 9 and a maximum score of 36 could be obtained. Higher scores indicate increased depression severity. The PHQ-9 demonstrates strong internal consistency, with a high Cronbach's alpha reported in .86 of studies, indicating excellent reliability across diverse cultural contexts. However, improvements are needed in reporting test-retest reliability (.24) and inter-rater reliability (.02) (Carroll et al., 2020).

Procedure

The ethical approval for this study was granted by the University of Amsterdam's Ethics Review Board before the data collection (FMG-11147_2024). The study was designed as a cross-sectional study and was conducted online in the form of a questionnaire that is a composite of multiple validated questionnaires divided over four subsections. The questionnaire was made available through the online application by Qualtrics. The total time for completing the entire questionnaire was estimated to be around 30 minutes total. Before presenting the questionnaires, participants were asked for their consent as well as several demographic questions. Next, participants completed the PHQ-9, GAD-7 and BSI in randomised order. This was followed by the EMNA questionnaire and the following questionnaires in randomised order: CTQ, DERS-SF and ECR. After the participant answered all the required questions, the participant was thanked for their participation on the final screen of the online questionnaire. They were also informed that their data remained anonymous. Finally, the participants could participate in a 50 euros raffle or gain 0.5 research credits in case they are students at the University of Amsterdam.

Analytic plan

All statistical analyses were coded and executed using R version 4.3.1. The analytical and data processing scripts were maintained in a private GitHub repository.

Correlations

To examine the relationships between the variables, Pearson's correlation coefficients were calculated, and scatterplots were generated to inspect the associations visually.

Checks for outliers

Outliers were identified using Studentized residuals, where any data point with an absolute Studentized residual greater than 3 was considered an outlier.

Statistical assumptions

The assumptions were validated for the following regression model with and without outliers: The encapsulated beliefs' strength proportion and DERS-SF non-acceptance score on the sum score of the PHQ-9 ($X, M \rightarrow Y$). These assumptions were examined in line with the principles of OLS regression as outlined by Hayes (2022). The normality of the residuals was assessed through both statistical and visual methods. The Shapiro-Wilks test was conducted to statistically evaluate normality, while a Q-Q plot was used for a visual inspection of the residuals. The assumption for homoscedasticity was evaluated by plotting the residuals against the predicted values in a scatter plot. The linearity of the relationships was assessed using Component-Component plus Residual plots (CCPRs). Finally, the multicollinearity assumption was tested using the Variance Inflation Factor (VIF).

Mediation analysis

Mediation analysis was conducted using the PROCESS macro developed by Hayes (2022) that is built for the version of R 4.3.1. The particular model selected was model 4, which corresponds to a simple mediation analysis. The independent variable was determined by using the believability of an encapsulated belief expressed as a proportion, while the mediator variable, the DERS-SF non-acceptance score, was obtained by summing all items in the non-acceptance subscale; likewise, the PHQ-9 score was computed by adding the individual responses to each PHQ-9 item. Due to the violation of the homoscedasticity assumption, a Heteroscedasticity-Consistent Standard Error estimator (HC3) was used to provide more robust standard error estimates. Bootstrapping samples were set to 5000 with an alpha of .05 and a seed of 1234 to ensure a similar random number basis between excluding and including outliers. Bootstrapped confidence intervals (95%) were computed to test the significance of the direct and indirect effects. The results of the mediation analysis were

interpreted according to the guidelines proposed by Hayes (2022), focusing on the total, direct, and indirect effects.

Sensitivity analysis

To investigate the impact of outliers on the mediation analysis, the mediation analysis was repeated with the exclusion of outliers.

Exploratory analysis

The variable used for the encapsulated belief strength construct for the main analysis is the degree of the current believability as measured in the EMNA questionnaire. To further examine the construct of an encapsulated belief, an additional mediation analysis was conducted on the negativity of an encapsulated belief that is also reported in the EMNA questionnaire. Prior to analysis, statistical assumptions are again tested. To address homoscedasticity violations, bootstrapping and HC3 estimators are employed in PROCESS. The mediation model was tested using the same statistical approach as in the main analysis.

Results

Preliminary analysis

Before conducting further statistical tests, the frequency distribution of each variable was examined (see Appendix A). Additionally, this section includes an assessment of correlations between key variables and tests of statistical assumptions to ensure the validity of subsequent analyses.

Correlations

The relationships among the independent variable, mediator and dependent variable were analyzed using a Pearson correlation coefficients table as shown in Table 1, as well as scatterplots that are shown in Figure 1 and Figure 2. As indicated by both the table and

figures, only PHQ-9 shows a significant correlation with DERS non-acceptance, while the other variables reveal insignificant associations.

Table 1

Pearson correlations of all variables

	PHQ-9	Encapsulated belief strength	DERS non-acceptance
PHQ-9	-	0.06	0.48*
Encapsulated belief strength	0.06	-	0.09
DERS non-acceptance	0.48*	0.09	-

*Note: * = Statistically significant effect relationship ($p < .05$)*

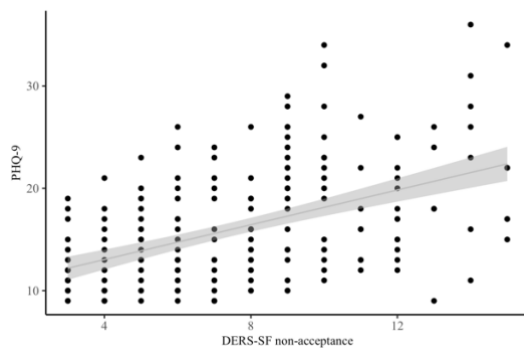


Figure 1. Scatterplot of DERS-SF non-acceptance scores on PHQ-9 scores.

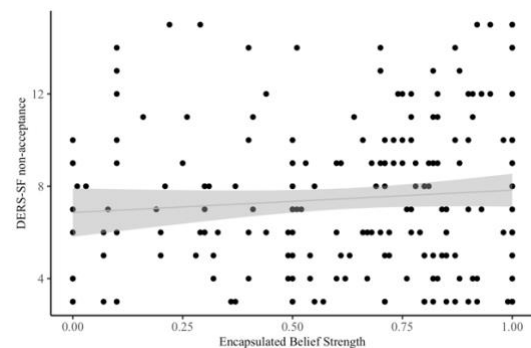


Figure 2. Scatterplot of Encapsulated belief strength on DERS-SF non-acceptance scores

Statistical assumptions

The assessment of normality with the Shapiro-Wilks test showed a violation for the regression model. The Q-Q plot also revealed some deviation from the line at both tails, particularly at the extremes. Homoscedasticity was also violated as it revealed a funnel shape

in the residual scatter plot. Both violations were addressed by applying bootstrapping in the PROCESS mediation analysis (Hayes, 2022) as well as applying HC3 estimators on the standard errors. The assumption of linearity was met by visual examination of the CCPRs for both regression models with and without outliers. Lastly, the VIF analysis revealed negligible multicollinearity among the predictors in the second regression model, both with and without outliers, with VIF values approximately equal to 1.

Mediation analysis

The analysis included the data of 211 participants. Figure 3 shows the results of the mediation analysis in the path model. The overall model explained 0.5% of the variance in depressive symptoms ($R^2 = .005$, $F = 1.18$, $p = .279$). The results showed that the total effect of encapsulated belief strength on PHQ-9 score was not statistically significant, $b = 1.34$, $\beta = 0.07$, $t = 1.09$, $se = 1.23$, $p = .279$, 95% bCI [-1.09, 3.77]. When the individual paths were examined, the strength of an encapsulated belief did not significantly predict the DERS-SF non-acceptance score, $b = 0.97$, $\beta = 0.09$, $t = 1.27$, $se = 0.79$, $p = .223$, 95% bCI [-0.59, 2.53]. However, the DERS-SF non-acceptance score significantly predicted the PHQ-9 score, $b = 0.85$, $\beta = 0.49$, $t = 7.43$, $se = 0.11$, $p < .001$, 95% bCI [0.62, 1.07], when controlling for encapsulated belief strength. The direct effect of encapsulated belief strength on PHQ-9 was not significant, $b = 0.52$, $\beta = 0.03$, $t = 0.47$, $se = 1.10$, $p = .636$, 95% bCI [-1.65, 2.69]. The indirect effect was also not statistically significant, $b = 0.82$, $se = 0.69$, bCI [-0.45, 2.23]. These results are not in line with the hypothesis that emotional non-acceptance partially mediates the relationship between encapsulated belief strength and depressive symptoms.

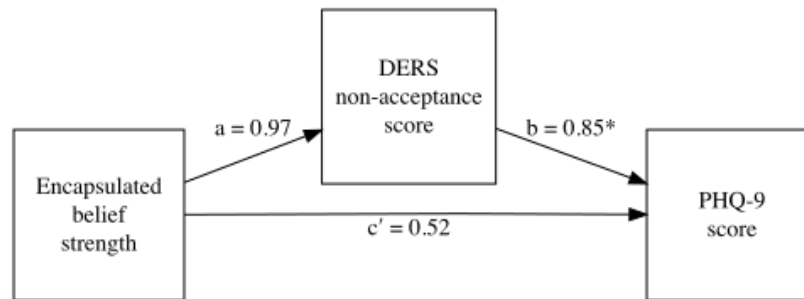


Figure 3. Results of PROCESS mediation analysis displayed in model path relations including outliers. The mediation analysis revealed a non-significant indirect effect ($a*b$ bootstrapped CI 95% [-0.45, 2.23])

* Reflects statistically significant effect relationship ($p < .05$)

Sensitivity analysis

As a result of a Studentized exclusion of outliers, the data of two participants were removed, leaving $N = 209$ in the final analysis. All Studentized residuals are plotted in Figure 4. The removal of outliers did not impact the tests of statistical assumptions, as both normality and homoscedasticity remained violated. The mediation analysis was conducted using the same configuration as the analysis that included outliers. Results remained robust, indicating that the analysis was not sensitive to their removal.

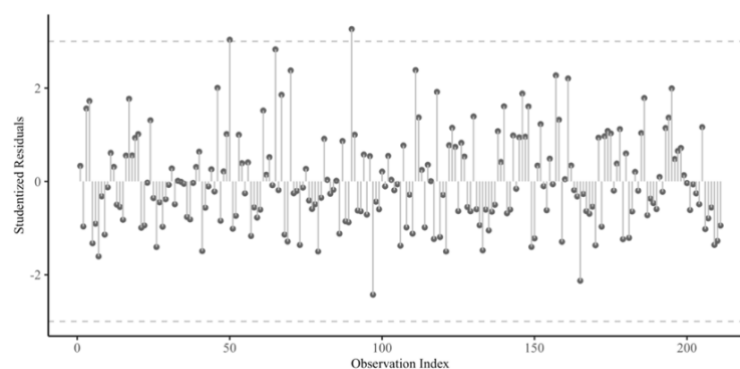


Figure 4. Plot of Studentized residual checks. Two datapoints were considered outliers.

Exploratory analysis

To explore the operationalization of encapsulated beliefs further, the same analysis was performed but this time using the negativity of an encapsulated belief, instead of the degree of believability. All statistical assumptions were tested, and normality and homoscedasticity were violated similarly to the main mediation analysis. Because of this, bootstrapping and HC3 estimators were again used to mitigate these violations in PROCESS. The analysis included 209 participants; two participants were removed due to missing data of encapsulated belief negativity. Figure 5 shows the results of the mediation analysis in the path model. The overall model explained 4.5% of the variance in depressive symptoms ($R^2 = .045$, $F = 10.09$, $p < .002$). The results showed a statistically significant total effect of encapsulated belief negativity on PHQ-9 score, $b = 4.91$, $\beta = 0.28$, $t = 4.58$, $se = 1.07$, $p < .001$, 95% bCI [2.79, 7.02]. When examining the individual paths, the negativity of an encapsulated belief did significantly predict the DERS-SF non-acceptance score, $b = 2.15$, $\beta = 0.21$, $t = 3.18$, $se = 0.68$, $p < .002$, 95% bCI [0.82, 3.49]. Additionally, the DERS-SF non-acceptance score did also significantly predict the PHQ-9 score, $b = 0.78$, $\beta = 0.46$, $t = 6.96$, $se = 0.11$, $p < .001$, 95% bCI [0.56, 1.01], when controlling for encapsulated belief negativity. The direct effect of encapsulated belief negativity on PHQ-9 was also significant, $b = 3.22$, $\beta = 0.18$, $t = 3.27$, $se = 0.98$, $p < .002$, 95% bCI [1.28, 5.16]. The indirect was also statistically significant, $b = 1.69$, $se = 0.61$, 95% bCI [0.59, 3.04]. These results suggest that the effect of an encapsulated belief's negativity predicts depressive symptoms and is partially mediated by emotional non-acceptance.

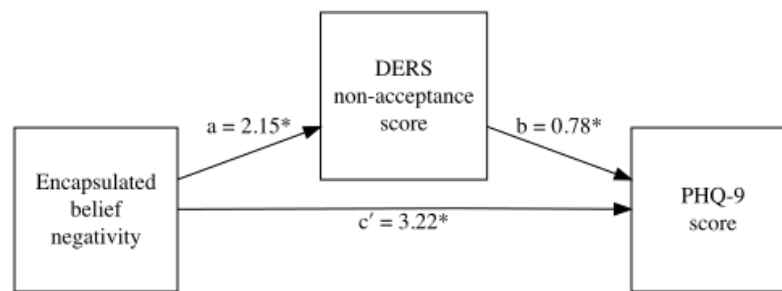


Figure 5. Results of PROCESS mediation analysis displayed in model path relations including outliers. The mediation analysis revealed a significant indirect effect ($a*b$ bootstrapped CI 95% [0.59, 3.04]) and a significant direct effect (c' bootstrapped CI 95% [1.28, 5.16])

* Reflects statistically significant effect relationship ($p < .05$)

Discussion

This study aimed to examine whether emotional non-acceptance partially mediates the relationship between the strength of an encapsulated belief from a negative experience and depressive symptoms. Based on previous research and the EST framework, partial mediation was expected. However, the findings challenge this assumption, suggesting that the degree to which individuals hold these beliefs with conviction may not be influencing emotional processing or depressive symptoms. These findings remained robust after the removal of outliers in the sensitivity analysis. Interestingly, the exploratory analyses revealed that when encapsulated belief strength was framed in terms of *negativity* rather than *believability*, a significant partial mediation effect emerged. This could suggest that the way an individual perceives the emotional valence of their beliefs may be more relevant to depressive symptoms than how strongly they regard these beliefs to be true.

Previous studies provided evidence to presume that there is an association between core beliefs and depressive symptoms, as shown by Osmo et al. (2018) and others. This

association was not found in the present study. A key distinction, however, is that those studies focussed on core beliefs in general while this study investigated the believability of a particular encapsulated belief related to a negative memory. Because of the focus on a specific memory, the sample population might have played a role in the reason why the relationship between the strength of an encapsulated belief and depressive symptoms was not found in the present study. Because it was a non-clinical sample, the particular memories, and therefore the encapsulated beliefs that the participants had, might not have been relevant or strong enough to correlate with depressive symptoms. Beck's cognitive model of depression emphasizes that beliefs related to *worthlessness* and *helplessness* are particularly central to depressive symptomatology. It is reasonable to assume that not all negative beliefs are equally relevant to depression and that only certain types of beliefs, such as those that challenge fundamental self-worth and autonomy, may strongly contribute to depressive symptoms. Since this study did not distinguish between different types of encapsulated beliefs, it remains unclear whether certain categories of beliefs are more strongly associated with emotional non-acceptance and depression than others.

Secondly, the findings for the relationship between emotional non-acceptance and depressive symptoms proved to be significant. Similarly to previous research from Bakhshaie et al. (2014) and Brockmeyer et al. (2012), which showed the same significant results. This is, however, not surprising as this relationship has been well established in the existing literature.

The relationship between encapsulated belief strength and emotional non-acceptance was also found to be non-significant. It was assumed that the rigidity of a belief might reflect the same underlying mechanism of cognitive rigidity as emotional non-acceptance. Therefore, non-acceptance could explain a similar stance in an encapsulated belief's believability. While this assumption seems reasonable, the two constructs are distinct, as

beliefs are cognitive in nature, whereas emotional regulation pertains to emotions. One could argue that the believability of a particular belief cannot be explained or associated with emotional non-acceptance if the belief itself is not inherently emotional. This distinction aligns with recent findings on cognitive restructuring and imagery rescripting (ImRs; Voncken et al., 2023), which suggest that belief modification strategies primarily influence the cognitive aspects of memory but may not directly alter emotional responses. The study demonstrates that while cognitive restructuring and ImRs effectively reduced the believability of encapsulated beliefs, neither method had a significant impact on the emotional connotation or distress associated with the memory.

The distinction between believability and negativity raises some questions about how the construct of encapsulated belief strength is represented. While the believability of an encapsulated belief is used as a treatment outcome measure in ImRs (Voncken et al., 2023), it has not been directly associated with depression nor emotional non-acceptance. There is some research however that did find reductions in distress and PTSD symptoms when the believability of an encapsulated belief is modified during ImRs treatment (Rameckers et al., 2024). Nevertheless, because the link between the strength of an encapsulated belief and emotional non-acceptance has not been researched before, the construct was further analysed in the exploratory analysis by replacing the *believability* of an encapsulated belief with the perceived *negativity* of the belief.

Interestingly, all relationships between the constructs and the mediation of emotional non-acceptance in the exploratory analysis proved to be significant when the negativity of an encapsulated belief was used. Its positive relation to emotional non-acceptance might be explained by the fact that how negatively an individual perceives their encapsulated belief may be more relevant for emotional processing difficulties than the believability. Supporting this, a study by Bailen et al. (2022) demonstrates a bidirectional relationship between

negative emotion and emotional nonacceptance, indicating that negative emotions and the rejection of those emotions reinforce each other over time. This suggests that the more negatively an individual experiences their emotional states, the greater their difficulties with accepting those emotions. The stark difference in results when framing encapsulated beliefs, also points to a limitation of the study. The lack of validated measures of the construct of encapsulated belief strength limits the studies ability to give conclusive evidence for it. However, it does press on the importance of how encapsulated beliefs are framed towards the participant. Future research should further explore the most reliable and meaningful way to assess encapsulated beliefs, considering whether negativity, believability, another construct or a composite is best representative of its strength.

Lastly, another limitation of this study is that the EMNA questionnaire was administered online. This part of the questionnaire was particularly important because it required participants to recall a negative emotional memory and provide their associated encapsulated belief. Unlike other questionnaires, this task required active memory recall, making it more demanding and time-consuming. The online format may have been less suitable, as participants could have perceived this task as overly time-consuming and chosen to fill it in quickly. Initial efforts to convert this questionnaire into a structured interview have been made and could be further explored in future studies.

This study aimed to deepen the understanding of depression by examining its relationship with beliefs and emotion regulation. While the primary analysis did not support the hypothesized mediation model, exploratory findings suggest that the perceived negativity of an encapsulated belief may be a stronger predictor of depressive symptoms through emotional non-acceptance than its believability. These results highlight the need for a more refined conceptualization and measurement of encapsulated beliefs, particularly in distinguishing their cognitive and affective components and their differential impact on

emotional processing. Given that this study was conducted in a non-clinical sample using an online questionnaire, future research should investigate these mechanisms in clinical populations and employ alternative assessment methods, such as structured interviews, to enhance measurement reliability. More broadly, a deeper understanding of encapsulated beliefs could inform therapeutic strategies such as ImRs. These findings highlight the discrepancy in understanding the cognitive-emotional mechanisms underlying psychopathology, suggesting the need for further exploration of how belief-based and emotion-focused approaches may contribute to interventions for depression.

References

- American Psychiatric Association. (2013). *Diagnostic and Statistical Manual of Mental Disorders*. American Psychiatric Association.
<https://doi.org/10.1176/appi.books.9780890425596>
- Arntz, A., & Weertman, A. (1999). *Treatment of childhood memories: theory and practice*.
- Bailen, N. H., Koval, P., Strube, M., Haslam, N., & Thompson, R. J. (2022). Negative Emotion and Nonacceptance of Emotion in Daily Life. *Emotion*, 22(5), 992–1003.
<https://doi.org/10.1037/emo0000898>
- Bakhshaie, J., Zvolensky, M. J., Brandt, C. P., Vujanovic, A. A., Goodwin, R., & Schmidt, N. B. (2014). The Role of Anxiety Sensitivity in the Relationship Between Emotional Non-Acceptance and Panic, Social Anxiety, and Depressive Symptoms Among Treatment-Seeking Daily Smokers. In *International Journal of Cognitive Therapy* (Vol. 7, Issue 2).
- Becerra, R., Naragon-Gainey, K., Gross, J. J., Ohan, J., & Preece, D. A. (2024). Beliefs about emotions: Latent structure and links with emotion regulation and psychopathology. *Journal of Affective Disorders Reports*, 16. <https://doi.org/10.1016/j.jadr.2024.100728>
- Beck, A. T. (1979). *Cognitive therapy and the emotional disorders*.
- Beck, A. T., & Bredemeier, K. (2016). A unified model of depression: Integrating clinical, cognitive, biological, and evolutionary perspectives. *Clinical Psychological Science*, 4(4), 596–619. <https://doi.org/10.1177/2167702616628523>
- Beck, J. S. (2011). *Cognitive Behavior therapy*.
- Brockmeyer, T., Bents, H., Holtforth, M. G., Pfeiffer, N., Herzog, W., & Friederich, H. C. (2012). Specific emotion regulation impairments in major depression and anorexia nervosa. *Psychiatry Research*, 200(2–3), 550–553.
<https://doi.org/10.1016/j.psychres.2012.07.009>

- Carroll, H. A., Hook, K., Perez, O. F. R., Denckla, C., Vince, C. C., Ghebrehiwet, S., Ando, K., Touma, M., Borba, C. P. C., Fricchione, G. L., & Henderson, D. C. (2020). Establishing reliability and validity for mental health screening instruments in resource-constrained settings: Systematic review of the PHQ-9 and key recommendations. *Psychiatry Research*, 291. <https://doi.org/10.1016/j.psychres.2020.113236>
- Cowan, H. R., McAdams, D. P., & Mittal, V. A. (2019). Core beliefs in healthy youth and youth at ultra high-risk for psychosis: Dimensionality and links to depression, anxiety, and attenuated psychotic symptoms. *Development and Psychopathology*, 31(1), 379–392. <https://doi.org/10.1017/S0954579417001912>
- Danasasmita, F. S., Pandia, V., Fitriana, E., Afriandi, I., Purba, F. D., Ichsan, A., Pradana, K., Santoso, A. H. S., Mardhiyah, F. S., & Engellia, R. (2024). Validity and reliability of the Difficulties in Emotion Regulation Scale Short Form in Indonesian non-clinical population. *Frontiers in Psychiatry*, 15. <https://doi.org/10.3389/fpsyt.2024.1380354>
- Dozois, D. J. A., & Rnic, K. (2015). Core beliefs and self-schematic structure in depression. In *Current Opinion in Psychology* (Vol. 4, pp. 98–103). Elsevier. <https://doi.org/10.1016/j.copsyc.2014.12.008>
- Ellis, A. (2003). Early theories and practices of rational emotive behavior therapy and how they have been augmented and revised during the last three decades. In *Inc. Journal of Rational-Emotive & Cognitive-Behavior Therapy* (Vol. 21). Human Sciences Press.
- Flynn, J. J., Hollenstein, T., & Mackey, A. (2010). The effect of suppressing and not accepting emotions on depressive symptoms: Is suppression different for men and women? *Personality and Individual Differences*, 49(6), 582–586. <https://doi.org/10.1016/j.paid.2010.05.022>
- Gratz, K. L., & Roemer, L. (2004). Multidimensional Assessment of Emotion Regulation and Dysregulation: Development, Factor Structure, and Initial Validation of the Difficulties

- in Emotion Regulation Scale 1. In *Journal of Psychopathology and Behavioral Assessment* (Vol. 26, Issue 1).
- Hayes, A. F. (2022). *Introduction to Mediation, Moderation, and Conditional Process Analysis* (3d Edition). Guilford Press.
- Kaufman, E. A., Xia, M., Fosco, G., Yaptangco, M., Skidmore, C. R., & Crowell, S. E. (2016). The Difficulties in Emotion Regulation Scale Short Form (DERS-SF): Validation and Replication in Adolescent and Adult Samples. *Journal of Psychopathology and Behavioral Assessment*, 38(3), 443–455.
<https://doi.org/10.1007/s10862-015-9529-3>
- Kraaij, V., Arensman, E., & Spinhoven, P. (2002). Negative Life Events and Depression in Elderly Persons: A Meta-Analysis. In *Journal of Gerontology* (Vol. 57, Issue 1).
<https://academic.oup.com/psychsocgerontology/article/57/1/P87/576198>
- Kroenke, K., Spitzer, R. L., & Williams, J. B. W. (1999). *The PHQ-9 Validity of a Brief Depression Severity Measure*.
- Leahy, R. L. (2022). *Emotional Schemas ScienceDirect*.
www.elsevier.com/locate/cabp Available online at www.sciencedirect.com
- Luchetti, M., & Sutin, A. R. (2016). Measuring the phenomenology of autobiographical memory: A short form of the Memory Experiences Questionnaire. *Memory*, 24(5), 592–602. <https://doi.org/10.1080/09658211.2015.1031679>
- McLaughlin, K. A., & Lambert, H. K. (2017). Child trauma exposure and psychopathology: mechanisms of risk and resilience. In *Current Opinion in Psychology* (Vol. 14, pp. 29–34). Elsevier B.V. <https://doi.org/10.1016/j.copsyc.2016.10.004>
- Nolen-Hoeksema, S. (2000). The Role of Rumination in Depressive Disorders and Mixed Anxiety/Depressive Symptoms. In *Journal of Abnormal Psychology* (Vol. 109, Issue 3).

- Osmo, F., Duran, V., Candidate, M., Wenzel, A., De Oliveira, I., Nepomuceno, S., Candidate, B., Madeira, M., & Menezes, I. (2018). The Negative Core Beliefs Inventory (NCBI): Development and Psychometric Properties. *Journal of Cognitive Psychotherapy*, 32(1).
- Philippe, F. L., Lecours, S., & Beaulieu-Pelletier, G. (2009). Resilience and positive emotions: Examining the role of emotional memories. *Journal of Personality*, 77(1), 139–176. <https://doi.org/10.1111/j.1467-6494.2008.00541.x>
- Qualtrics. (2025). *Qualtrics*. <https://www.qualtrics.com/>
- Rameckers, S. A., van Emmerik, A. A. P., Boterhoven de Haan, K., Kousemaker, M., Fassbinder, E., Lee, C. W., Meewisse, M., Menninga, S., Rijkeboer, M., Schaich, A., & Arntz, A. (2024). The working mechanisms of imagery rescripting and eye movement desensitization and reprocessing: Findings from a randomised controlled trial. *Behaviour Research and Therapy*, 175. <https://doi.org/10.1016/j.brat.2024.104492>
- Shoemann, A. M., Boulton, A. J., & Short, S. D. (2024). *Monte Carlo Power Analysis for Indirect Effects*. https://schoemanna.shinyapps.io/mc_power_med/
- Stange, J. P., Alloy, L. B., & Fresco, D. M. (2017). Inflexibility as a Vulnerability to Depression: A Systematic Qualitative Review. In *Clinical Psychology: Science and Practice* (Vol. 24, Issue 3, pp. 245–276). Blackwell Publishing Inc. <https://doi.org/10.1111/cpsp.12201>
- Voncken, M. J., Janssen, I., Keijsers, G. P. J., & Dibbets, P. (2023). Should cognitive restructuring precede imagery rescripting? An experimental pilot study. *Journal of Behavior Therapy and Experimental Psychiatry*, 78. <https://doi.org/10.1016/j.jbtep.2022.101800>
- World Health Organization. (2021, August 13). *Depressive disorder (depression)*. <https://www.who.int/news-room/fact-sheets/detail/depression>

Yoon, S., Dang, V., Mertz, J., & Rottenberg, J. (2018). Are attitudes towards emotions associated with depression? A Conceptual and meta-analytic review. In *Journal of Affective Disorders* (Vol. 232, pp. 329–340). Elsevier B.V.
<https://doi.org/10.1016/j.jad.2018.02.009>

Appendix A

